Name.....

Signature.....**Candidates Personal Number**.....

545/1.

CHEMISTRY.

Paper 1

Aug /2025.

2 hours

CHEMISTRY DEPARTMENT

New Lower Secondary Chemistry Examination

Uganda Certificate of Education

Internal mock set 10

Paper 1

2 hours

INSTRUCTIONS:

This paper consists of **two** *sections* **A** *and* **B**. *It has* **six** *examination items.*

Section **A** *has* **one compulsory** items.

Section **B** *has* **two parts, I** and **II**. *Answer* **one** item from each part.. Answer **four** items in all.

Answers to section **A must** *be written in the spaces provided while those of section* **B must** *be written on the provided answer sheets.*

Any additional item(s) answered will **not** *be scored.*

Where necessary use C = 12, H = 1

SECTION A.

Answer **all** *items in this section in the spaces provided*

Item 1

At the end of year party, a variety of beverages were served. John chose one type which, after he drank it, initially made him feel excited and relaxed. However, his movements later became unconditioned and lacked self control. His friends, who were taking other type of beverage that didn't have the same effect on them, repeatedly warned him to stop consuming any more of that beverage.

Task

As a learner of chemistry,

(a) State the **category** of the product John took.

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(b) Suggest any other use of the active ingredient contained in the product John took.

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(c) explain how excess consumption of such products John used can have serious consequences.

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(d) Compare the products served at the party

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Item 2.

In an effort to promote energy and environmental sustainability, the government has recommended a specific hydrocarbon product for public use. The label on the product indicates 74.98% carbon by mass and 1 mole of product weighs 16.01 grams. However, the government must first educate the public about the product and develop strategies for its environmentally responsible use. You have been invited to one of the educational workshops.

Task: As a Chemistry student,

(a) explain the category of the **structure** possessed by the product.

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(b) Determine the molecular formula of the product.

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(c) Suggest the properties of the product that make it useful in every day life.

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(d) advise on the environmental impact associated with use of the product.

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SECTION B

Part I (*Attempt one item from this section)*

Item 3 copper is a versatile metal used for electrical wiring, making cook ware , electronics , brass and bronze alloys among others .To increase on quantities of the metal on market needed , the government has cleared an investor to set up a copper extraction plant in kilembe ,Kasese Uganda. However ,the locals are concerned about its impact and are seeking guidance on how they will benefit from it's operations. The officials have organized a workshop to educate the locals. You have been invited to attend.

Task. *As* a chemistry student, make a write up you will use during the presentation.

Item 4

Large quantities of sulphuric acid are used as electrolyte in car batteries ,in production of ammonium sulphate fertilizer , dyes and paint. Jinja city officials have cleared an investor to set up an Sulphuric acid manufacturing plant. However, the locals are concerned about its impact to the environment and how they will socially benefit from the plant. The officials have organized a workshop at the city hall. You have been invited to address the locals.

Task. Prepare a write up message you will use.

Part II (*Attempt* **one** *item* from *part*)

Item 5 The dependence on **sedimentary rocks** for valuable minerals in vast areas of eastern Uganda has led to signifant environmental impact particularly to nearby water sources . In response, the regional environmental officials have organized a series of public workshops aimed at raising awareness among the locals the importance of balancing resource use with crucial need for protecting their environment. You have been invited to attend and address the locals.

Task. Prepare a write up message you will deliver.

Item 6 The Uganda's National Development Plans (NDPs) Vision 2040 emphasizes the need for adoption and mitigation strategies to protect Uganda from climate change impact. One of the initiatives is to educate the public about **sustainable use of wood charcoal** and **natural gas** for energy. Officials from Ministry of Environment have organized radio education programs to address the public . You have been invited to one of the programs.

Task. Write a write up presentation you will use during the presentation.

End.